

A Comparison of Decision Tree Ensembles and Naïve Bayes

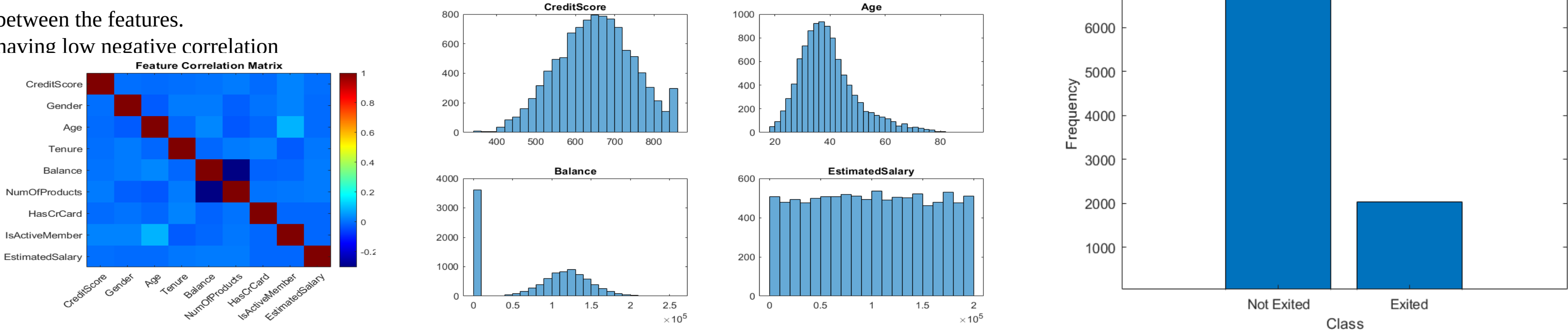
Description and Motivation

- Building Decision Tree Ensembles and Naïve Bayes supervised learning algorithms for classification task.
- Analyzing and comparing the performance of the 2 models for classifying whether a customer left the bank or is an existing customer

Exploratory Analysis

- The dataset, known as "Churn Modelling," available on Kaggle, is used for categorizing bank customers into groups based on whether they are likely to leave or stay with the bank.
- The dataset consists of 14 columns and 10000 rows; we will be using 10 columns as the features and 1 target (Exit) which is binary
- 8 features are numerical and 2 categorical. The 2 categorical predictors were turned into numerical ones.
- .The class imbalance problem is present as the Exited (1) customer’s number is 2037, whereas the not Exited(0) customer number is 7963.
- We also created histograms for the distribution of some of the features which could be rel
- The correlation heatmap shows the correlation between the features.
- There are no significant correlation some even having low negative correlation
- Some of the features exhibit Gaussian

distribution, whereas others are more uniformly distributed. Interestingly, a lot of people had 0 Balance.



Decision Tree Ensembles

- Ensemble learning technique leveraging multiple decision trees to improve predictive performance.
- Utilizes bagging by training individual trees on bootstrap samples of the dataset, enhancing robustness against overfitting.
- Model diversity is achieved through the independent training of trees, reducing the variance of the collective predictions (Khan et al., 2019).
- Aggregation method for classification is typically majority voting, where the most common label among all trees is chosen.
- Employs the 'fitensemble' function with 'Method' set to 'Bag', indicating the use of bagging.
- Hyperparameter tuning is integrated, optimizing 'MaxNumSplits', 'MinLeafSize', and 'MinParentSize' via a grid search to refine individual trees.
- The ensemble's final prediction is designed to be more accurate and stable than that of a single decision tree.

Advantages

- Offer strong prediction accuracy
- Combining multiple models to decrease variance, more reliable
- Good at handling noises in the data
- Effective for classification problems
- Leverages computational resources efficiently as the trees are trained in parallel.models

Disadvantages

- Can be complex
- Requires high computational resources for training and prediction
- Introduces risk of bias if not well tuned.
- Time consuming for large datasets
- Challenging updating models for new data, would require more time and be hard to tweak.

Naïve Bayes

- A probabilistic classifier based on Bayes' theorem with the assumption of feature independence within each class.
- Efficient for large datasets, often with a fast training phase, due to its simplicity and use of conditional probability.
- Particularly effective for text classification tasks.
- Utilizes a straightforward approach that calculates the probability of each class and the conditional probability of each class given each input value.
- Typically involves less computational resources compared to complex models.
- Can perform well with a small amount of training data if the assumption of independent features holds.
- Hyperparameters in Naive Bayes are primarily the prior probabilities of the classes and the distributions assumed for the features.
- Naive Bayes can yield a robust baseline model that performs adequately across various application domains.

Advantages

- Quick training and prediction times
- Easy to implement and understand
- High performance on classification tasks
- Works well with high dimensional data
- Few hyperparameters

Disadvantages

- Relies on independence of the features
- Could produce less reliable probability estimates (Lowd & Domingos, 2005)
- Poor performance with complex featire relationships
- Requires data distribution assumptions for continuous data (Gaussian, Bernoulli, etc).

Hypothesis statement:

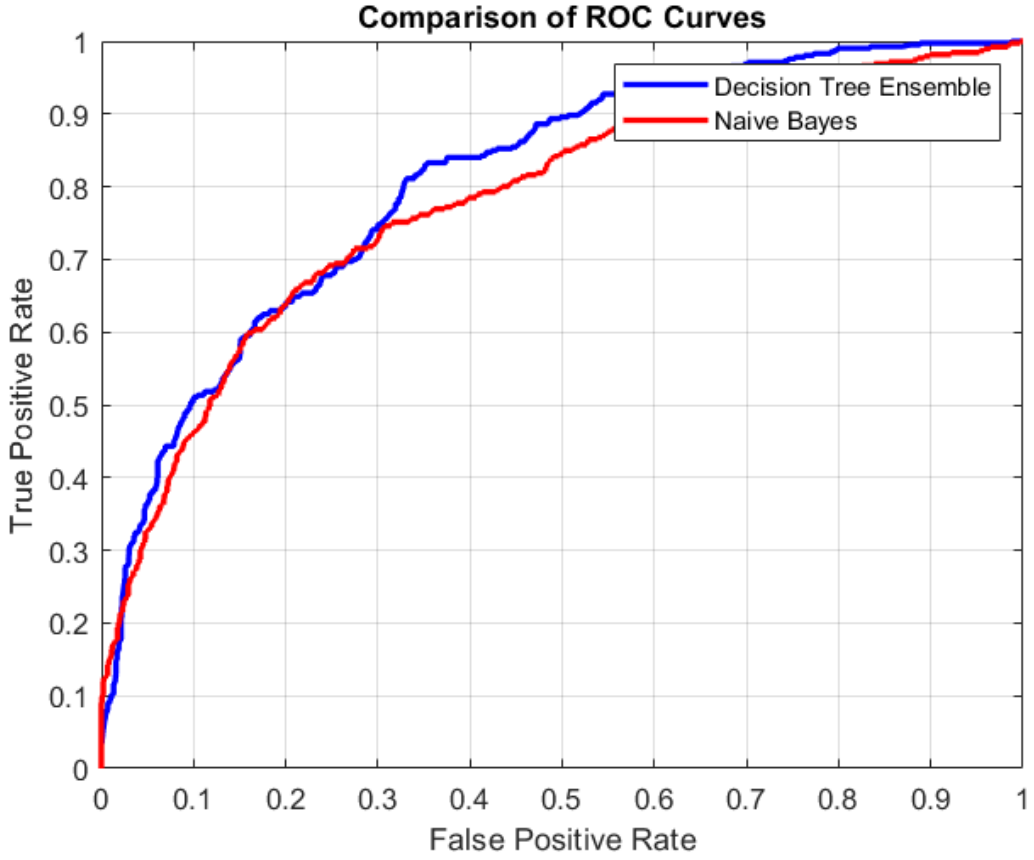
- Decision Tree Ensembles will require much longer training and testing time.
- There is an imbalance of data which would skew our results.
- The Naïve Bayes could perform better than the Decision Tree Ensembles (Yadav et al., 2019).

Choice of Parameters

- All of the features were used, where the categorical data was turned into numerical, as there is no correlation between the features.
- Divided the training and test by proportions of 80/20
- Used SMOTE since there was a high class imbalance
- Employing grid search for decision tree ensemble for MaxNumSplits, MinLeafSize, MinParentSize.
- Number of cycles for the ensemble chosen 100.
- Evaluated the 2 models by Accuracy, Precision, Recall, F1-Score and AUC.

Experimental Results

	Accuracy	Precisio n	Recall	F1- Score	AUC
NB	57.2%	0.93	0.51	0.636	0.78
DTE	76.6%	0.90	0.79	0.84	0.81



Methodology

- In this methodology, two machine learning models, Naive Bayes and Decision Tree Ensembles, are compared using a churn prediction dataset. The process begins with data loading and preprocessing, including conversion of categorical variables into numerical formats. Relevant features are selected for analysis, and the data is split into training and testing sets with 80/20 proportions. To address class imbalance, the Synthetic Minority Over-sampling Technique (SMOTE) is employed, enhancing the dataset's representativeness. Each model is then trained: Naive Bayes on a standard dataset and Decision Tree Ensembles with hyperparameter tuning. The models are evaluated based on accuracy, precision, recall, F1-score, and AUC, and their performance is visually compared using ROC curves. This approach provides a comprehensive comparison, highlighting the strengths and weaknesses of each model in predicting customer exit.

Analysis and Critical Evaluation

- Accuracy:** The Naive Bayes model shows a moderate accuracy of 57.20%, which is significantly lower compared to the Decision Tree Ensemble's accuracy of 76.6%. This suggests that DTE has learned the training data more effectively.
- Precision:** NB's precision is slightly higher at 0.93, compared to DTE's 0.90, suggesting that NB is marginally better at predicting true positives accurately.
- Recall:** DTE exhibits a considerably better recall of 0.79 against NB's 0.51, indicating DTE's superior capability in identifying actual churn cases.
- F1 Score:** With an F1 score of 0.84, DTE outperforms NB's 0.636, implying a more balanced performance between precision and recall.
- AUC:** The AUC for DTE stands at 0.8104, slightly higher than NB's AUC of 0.78. Although both models demonstrate a good ability to differentiate between classes, DTE holds a small edge.
- Model Training and Prediction Time:** Naive Bayes is much faster due to its simpler calculations, whereas Decision Tree Ensembles require more computational resources due to the complexity of multiple trees.
- Feature Sensibility:** All of the variables have been used for both NB and DTE making it more comprehensive
- Predictive Power:** DTE shows higher predictive power across all metrics, suggesting it is better suited for this particular dataset and task.
- These results suggest that DTE's approach to handling feature interactions and class imbalances is more effective for this dataset. However, the complexity and potentially lower interpretability of the DTE model should be considered in practical applications.

Lessons learned and Future Work:

- Use of SMOTE is very important as it balances the classes which makes it more fair
- The DTE takes very long time and thus much need more computational power
- Optimization for Naïve Bayes using kernel
- Cross-Validation Techniques use in the future
- Using more comprehensive approach for NB and possibly use of Random Forest rather than DTE
- Focus on enhancing NB's recall and exploring more advanced ensemble techniques or feature engineering to further improve the DTE model.

References:

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